

Name: _____ Date: _____

Period: _____

1. Complete the difference table and identify the best model type.

x	y	Δy	$\Delta^2 y$
0	5	—	—
1	9	4	—
2	17	8	4
3	29	12	4
4	45	16	4

The **second** differences are constant, so the best model type is **quadratic** (degree **2**).

2. A shipping company charges a flat rate of \$8 for packages up to 5 lb, then adds \$1.50 per pound above 5 lb. Circle the best model type and give one reason.
- A. Linear B. Quadratic C. Cubic D. Piecewise-defined ← **correct**

One reason: **The pricing rule changes at 5 lb, so different formulas apply on different intervals.**

3. A quadratic function models the area of a rectangular garden with one fixed wall (using 60 m of fencing for three sides). State one **domain restriction** and one **model assumption**.

Domain restriction: **$0 < w < 30$ where w is the width (width must be positive and less than half the available fencing).**

Model assumption: **The garden is a perfect rectangle and exactly 60 m of fencing is used on the three open sides.**